# Vitamin D Deficiency Detection Using Machine Learning


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# Contents

# Abstract


Vitamin D deficiency (VDD) is a major global health issue, highlighting the need for effective, easy, and non-invasive prediction methods. This study collected data from 250 participants aged 18 to 30 in Visakhapatnam, Hyderabad, and Maharashtra. The data included various demographic and lifestyle factors such as age, gender, weight, height, body mass index, waist circumference, body fat percentage, bone mass, exercise habits, sunlight exposure, milk consumption, and vitamin D supplementation.

The goal of this research is to predict VDD severity using different machine learning algorithms. Key metrics such as precision, recall, F1-score, accuracy, and AUC-ROC will be used to assess the performance of algorithms like K-Nearest Neighbors, Decision Trees, Random Forest, and Support Vector Machines. The study will also investigate a hybrid model that combines Convolutional Neural Networks and ResNet for better prediction accuracy.

Additionally, the research will analyze how lifestyle and environmental factors affect Vitamin D levels to identify key predictors of VDD severity. By employing SHAP and LIME techniques, the study aims to enhance the interpretability of the models. Ultimately, this work seeks to contribute to public health by providing tools for assessing VDD and raising awareness of lifestyle changes that can help maintain healthy Vitamin D levels.

# Introduction

Vitamin D Deficiency (VDD) affects nearly a billion people worldwide, posing serious health risks and linking to autoimmune diseases such as cardiovascular issues, diabetes, and certain cancers. Traditional methods of diagnosing VDD are often complex and time-consuming, leading to overlooked cases. This study utilizes advanced machine learning techniques, specifically a hybrid CNN and ResNet model, to predict VDD severity and classify cases into categories of deficiency, insufficiency, and sufficiency. Machine learning has become crucial in healthcare for its efficiency and ability to analyze complex datasets, making it an ideal choice for VDD prediction. Unlike prior studies that relied on smaller datasets and traditional models, this research incorporates a hybrid CNN and ResNet model that captures intricate data patterns for enhanced accuracy. Additionally, this study evaluates various algorithms—such as Linear Regression, Decision Trees, Random Forest, and others—using performance metrics like precision, recall, F1-score, and ROC curves. To improve model interpretability, Explainable AI tools such as SHAP and LIME are used, identifying key factors that influence VDD severity. This research contributes to public health efforts by providing an accurate, accessible diagnostic tool and raising awareness of lifestyle factors that support optimal Vitamin D levels.

# Literature Review

| Title | Authors and Year | Methodology Used | Advantages | Limitations |
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| Vitamin D Deficiency Detection: A Novel Ensemble Approach with Interpretability Insights | Md Fahim,2024 | The system uses CatBoost and LightGBM models to predict Vitamin D deficiency with 96% accuracy and explains the important lifestyle factors using SHAP. | The approach effectively predicts Vitamin D deficiency with high accuracy and offers valuable insights into lifestyle factors, improving healthcare decisions and patient outcomes. | The study may be impacted by potential dataset biases affecting generalizability, a need for further clinical validation, ethical considerations, and possible resource constraints for real-time monitoring deployment. |
| Proposing an Effective Deep Learning Model for Vitamin D Deficiency Diagnosis | Asad Khattak,2024 | Utilizes convolutional neural networks (CNNs) to classify Vitamin D deficiency based on pre-processed patient lifestyle and demographic data. | The deep learning model offers improved accuracy in classifying Vitamin D deficiency compared to traditional methods, allowing for more reliable risk assessments and targeted health interventions. | The model's performance is limited by its reliance on a dataset that lacks direct clinical or biochemical data, which may affect generalizability and interpretability. |
| A Novel Approach for Prediction of Vitamin D Status Using Support Vector Regression | Shuyu Guo,2013 | The study employed Random Forest (RF) and Radial Basis Function Support Vector Regression (RBF SVR) algorithms for modeling vitamin D levels. | The study demonstrates the effectiveness of machine learning algorithms in accurately modeling Vitamin D levels, potentially improving screening and management for individuals at risk of deficiency. | The study relied on a single validation dataset, struggled to predict extreme vitamin D levels, and may not accurately identify high-risk groups. |

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| A Predictive Performance Analysis of Vitamin D Deficiency Severity Using Machine Learning Methods | G Sambasivam,2020 | Multiple machine learning models were tested for predicting Vitamin D deficiency, with Random Forest achieving the highest accuracy of 96% after data cleaning and feature selection. | The study provides a reliable model for predicting Vitamin D deficiency severity in young adults, which can guide targeted interventions and improve health outcomes in this demographic. | The dataset's focus on 18-21-year-old college students and its location-specific results may limit the applicability and validation of findings across different age groups and climates. |
| Leveraging Machine Learning to Evaluate Factors Influencing Vitamin D Insufficiency in SLE Patients | Mrinal Saha,2023 | Data was collected from 50 SLE patients on demographics and vitamin D levels, and analyzed using Linear Regression and Random Forest models to determine important factors. | The study offers insights into the factors affecting Vitamin D levels in SLE patients, potentially guiding personalized treatment plans and improving patient management strategies. | The small sample size and the exclusion of significant factors, such as corticosteroid therapy and complement levels, limit the comprehensiveness and reliability of the findings. |

# Requirement Analysis

## *Functional Requirements*

**Algorithm Implementation:**

Random Forest (Bagging)

LightGBM & CatBoost (Boosting)

K-Nearest Neighbors (KNN) Multi-Layer Perceptron (MLP)

**Model Performance Evaluation:**

Accuracy

Precision Recall

F1-Score

AUC (Area Under Curve)

**Explainable AI (XAI) Techniques:**

SHAP (Shapley Additive Explanations)

LIME (Local Interpretable Model-Agnostic Explanations)

## Non-Functional Requirements

**preprocessing of Datasets:**

Models should handle datasets efficiently.

**Efficiency and Memory Usage:**

Prioritize low memory usage, especially for LightGBM.

**User-Friendly Explainability:**

Design XAI outputs to be interpretable by non-technical users.

**Inputs and Outputs:**

Inputs: Structured data (CSV, databases) with categorical and numerical variables.

Outputs: Model performance metrics and explainable decision insights.

# Tools/Technologies

## Software Requirements

- **Programming Language:**
    - **Python (3.8 or above):** Easy to read and widely used in data science.
- **Libraries:**
    - **Scikit-learn:** For machine learning algorithms and evaluation.
    - **XGBoost, LightGBM, CatBoost:** Efficient libraries for handling large datasets.
    - **TensorFlow/Keras:** For building neural networks, with support for GPU acceleration.
    - **SHAP and LIME:** Tools for explaining model predictions.
    - **Pandas and NumPy:** For data manipulation and numerical operations.
    - **Matplotlib and Seaborn:** For data visualization.
- **Development Environment:**
    - **Jupyter Notebook/Anaconda:** For interactive coding and managing libraries.
    - **Git:** For version control and collaboration.

## Hardware Requirements

1. **Intel i5 or higher or AMD Ryzen 5 or higher:** Ensures smooth processing for data processing and model training.
2. **Parallel Processing:** Multi-core processors or CPUs with multiple threads are recommended for enhanced performance with libraries like scikit-learn and TensorFlow.

**1.RAM:**

1. **16 GB or more:** Necessary for handling large datasets without crashes and enabling faster model training.
2. **Expandable RAM:** Preferred for future upgrades to handle even larger datasets or more intensive models.

**2.Storage:**

1. **512 GB SSD or higher:** Fast storage reduces latency in data processing and library loading.
2. **Cloud Storage Option:** Using cloud storage (e.g., AWS S3 or Google Cloud Storage) can optimize local storage and provide scalable access to large datasets.

# Conclusion

In conclusion, our project aims to develop a predictive model to classify individuals into four categories of vitamin D levels: sufficient, insufficient, deficiency, and severe deficiency. This is crucial given the widespread prevalence of vitamin D deficiency and its impact on public health. Early detection can lead to timely interventions, improving bone health and overall well-being.

We use a non-invasive approach with machine learning techniques, specifically a hybrid CNN-ResNet model, to enhance prediction accuracy. The study analyzes a diverse dataset from 250 samples aged 18-30, focusing on various lifestyle factors like age, BMI, exercise, and sunlight exposure.

Performance metrics such as precision, recall, and F1 score are key to evaluating our models. Explainable AI tools like SHAP and LIME improve the model's interpretability, helping healthcare professionals make informed decisions. The expected outcomes include a valuable tool for personalized healthcare recommendations, enabling early identification of at-risk individuals and supporting preventive measures. Ultimately, this project seeks to improve health outcomes by providing targeted interventions for vitamin D deficiency across diverse populations.

# References

[1] M. F. Ul Islam, M. Hasan, M. T. Rahman and A. Chakrabarty, "Vitamin D Deficiency Detection: A Novel Ensemble Approach with Interpretability Insights," 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), Dhaka, Bangladesh, 2024, pp. 137-142, doi: 10.1109/ICEEICT62016.2024.10534371.

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# Problem Identification And Objectives

Traditional diagnostic methods for Vitamin D deficiency (VDD) rely on invasive and costly biochemical assessments, which can lead to missed cases, especially in resource-limited settings. This research utilizes machine learning (ML) to provide a non-invasive alternative, employing a hybrid model of Convolutional Neural Networks (CNN) and ResNet to improve predictive accuracy by analyzing factors like age, BMI, and lifestyle habits.

ML's ability to analyze large datasets is increasingly valuable in healthcare for predicting conditions like VDD. This study compares the CNN-ResNet hybrid model with various ML algorithms, including Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting Classifier, to determine optimal performance based on Precision, Recall, F1-score, and ROC curves.

To enhance model interpretability, Explainable AI techniques like SHAP and LIME will be used to identify key features influencing VDD severity. Ultimately, this research aims to deliver an accurate, scalable diagnostic tool for VDD assessment, addressing public health needs and facilitating more accessible interventions.

## Objectives

1.**Develop ML Models for VDD Prediction**: Implement a range of machine learning classifiers—such as K-Nearest Neighbor (KNN), Decision Trees (DT), Random Forest (RF), AdaBoost, and more—to enhance prediction robustness across different population groups.

2.**Implement a Hybrid CNN-ResNet Model**: Use a hybrid deep learning approach combining CNN and ResNet for deep feature extraction and pattern recognition, aiming to improve prediction quality by capturing complex data interactions.

3.**Performance Evaluation**: Measure model effectiveness using metrics like Accuracy, Precision, Recall, F1-score, and ROC-AUC, to ensure high sensitivity and specificity in VDD predictions.

4.**Model Interpretability**: Integrate SHAP for feature interpretability, helping to highlight influential predictors such as lifestyle and demographic factors, thus supporting healthcare decision-making.

5.**Comparison with Traditional Models**: Evaluate the hybrid CNN-ResNet against traditional methods to demonstrate its efficiency and predictive accuracy, making it a potential non-invasive clinical tool.

6.**Develop a Scalable VDD Screening Tool**: Provide a scalable, accurate VDD detection solution that healthcare providers can use for accessible, timely interventions across varied clinical and community settings.

# Existing System

Predictive models for Vitamin D deficiency (VDD) typically use ensemble machine learning classifiers. These models combine several algorithms to improve accuracy, reliability, and robustness. Common ensemble methods like Random Forest (RF), Gradient Boosting (GB), and CatBoost analyze various input features, including demographic, lifestyle, and health factors such as BMI, sunlight exposure, and dietary habits. This allows for a non-invasive way to assess VDD risk using easily accessible patient data.

Ensemble models often perform well across large populations, helping to identify high-risk individuals who may need further biochemical tests or preventive treatments. Many studies report high accuracy rates—often above 90%—for VDD classification with models like RF and GB, thanks to their effective feature management and data correlation. These models have become benchmarks for non-invasive VDD detection, improving screening processes and enabling early intervention, especially in low-resource settings.

Additionally, techniques like SHAP (SHapley Additive exPlanations) are useful for understanding model predictions and the importance of different features. They help illustrate how factors like sunlight exposure or BMI influence the final predictions, which is important for patient education and planning interventions.

## Technical Limitations and Drawbacks

**Limited Generalizability**: Current VDD models may not work well for all populations due to differences in environment, diet, and genetics. This requires retraining with local data, which can take a lot of resources.

**Integration Challenges**: Many models are standalone and hard to connect with electronic health records, limiting their use in real-time clinical settings.

**Dependency on Structured Data**: These models require well-organized data to make accurate predictions. If data is missing or not aligned, it can lead to mistakes, making them less effective in clinics with outdated systems.

**Computational Constraints**: While these models need less computing power than complex ones, they still require some resources that might not be available in low-resource settings.

**Real-Time Application Challenges**: Many models process data in batches, making them less useful in situations where quick decisions are necessary.

**Accuracy Variation in Extreme Cases**: These models may not perform well for patients with unique health issues or complex conditions, which need special diagnostic criteria.

# Proposed System

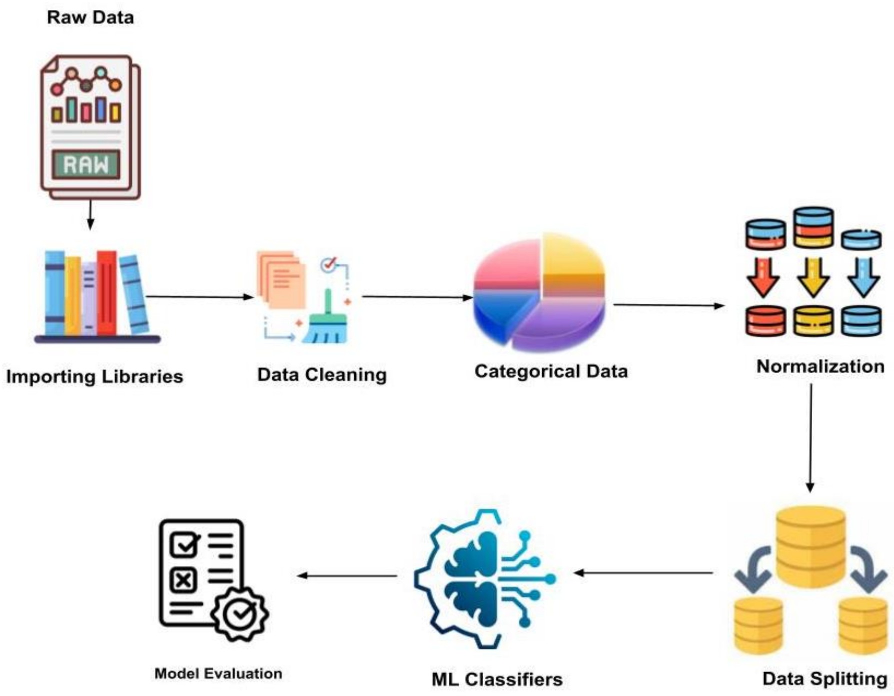


Fig. Proposed System

The Vitamin D Deficiency Detection System is an innovative machine learning solution designed to evaluate an individual's risk of Vitamin D deficiency based on their lifestyle choices. By analyzing data related to dietary habits, sunlight exposure, physical activity, and other relevant lifestyle factors, this system delivers precise assessments of Vitamin D deficiency risk directly to healthcare professionals, eliminating the need for clinical notes or laboratory test results.

Employing Explainable AI techniques, such as SHAP and LIME, the model ensures that its predictions are interpretable. This interpretability feature enhances transparency by providing healthcare providers with clear insights into the lifestyle factors contributing to deficiency risk, fostering trust and facilitating informed decision-making. Additionally, the system is built for real-time adaptability, allowing it to incorporate the latest research findings regarding lifestyle influences on Vitamin D. As a result, its predictions remain accurate and relevant over time, while its scalable design enables deployment across various healthcare settings.

This Vitamin D Deficiency Detection System represents a practical and proactive approach to preventive healthcare, focusing on lifestyle data to offer actionable insights that can help mitigate the risk of Vitamin D deficiency.

# System Architecture

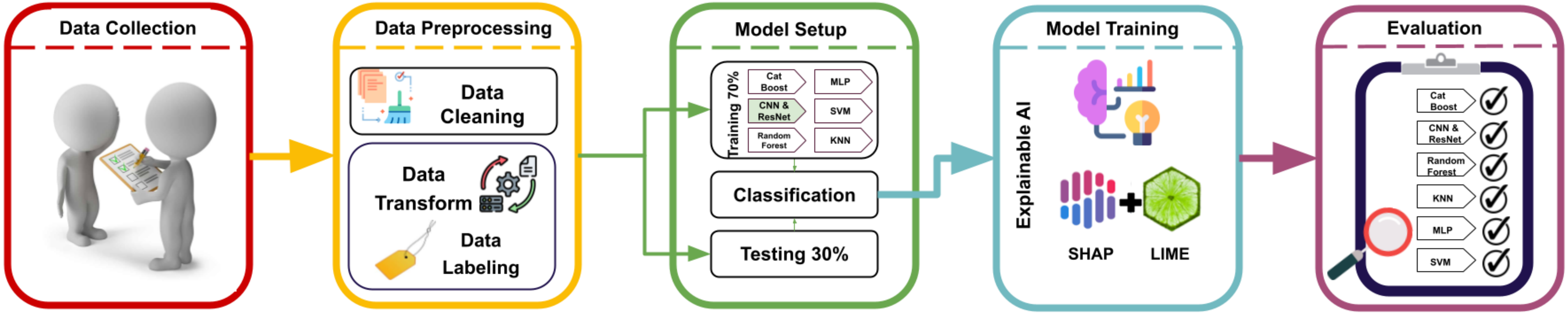


**Fig. System Architecture**

**The Vitamin D Deficiency (VDD) severity prediction system is being developed in three main stages:**

**1.Data Acquisition**: We have collected a dataset of 250 samples that focuses on important factors like age, body mass index (BMI), exercise habits, sunlight exposure, and milk intake. This data was gathered from regions in Maharashtra, Telangana, and Andhra Pradesh.

**2.Data Preprocessing**: In this phase, we clean the data by filling in missing values and detecting outliers to ensure the dataset is accurate. We standardize the data for consistency and prepare it for modeling by normalizing continuous variables and encoding categorical ones. We also balance the dataset to handle any imbalances between different classes.

**3.Machine Learning Model Implementation**: We are testing a hybrid model that combines Convolutional Neural Networks (CNN) and ResNet, alongside other models like K-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), LightGBM, CatBoost (CB), Bagging Classifier (BC), Gradient Boosting (GB), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) to predict the severity of VDD. The models are currently being fine-tuned, and early results suggest that the hybrid CNN-ResNet model performs well. We evaluate all models based on metrics such as Precision, Recall, F1-Score, and AUC.

**4.Explainable AI**: To make the model's predictions easier to understand, we are using tools like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). These tools help us see which factors are most important in the predictions, making the system more transparent and trustworthy.

# Tools/Technologies

## Software Requirements

•**Programming Language:**

- **Python (3.8 or above):** Easy to read and widely used in data science.

•**Libraries:**

- **Scikit-learn:** For machine learning algorithms and evaluation.
- **XGBoost, LightGBM, CatBoost:** Efficient libraries for handling large datasets.
- **TensorFlow/Keras:** For building neural networks, with support for GPU acceleration.
- **SHAP and LIME:** Tools for explaining model predictions.
- **Pandas and NumPy:** For data manipulation and numerical operations.
- **Matplotlib and Seaborn:** For data visualization.

•**Development Environment:**

- **Jupyter Notebook/Anaconda:** For interactive coding and managing libraries.
- **Git:** For version control and collaboration.

## Hardware Requirements

1. **Intel i5 or higher or AMD Ryzen 5 or higher:** Ensures smooth processing for data processing and model training.
2. **Parallel Processing:** Multi-core processors or CPUs with multiple threads are recommended for enhanced performance with libraries like scikit-learn and TensorFlow.

**1.RAM:**

1. **16 GB or more:** Necessary for handling large datasets without crashes and enabling faster model training.
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**2.Storage:**

1. **512 GB SSD or higher:** Fast storage reduces latency in data processing and library loading.
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# Conclusion

In conclusion, our project aims to develop a predictive model to classify individuals into four categories of vitamin D levels: sufficient, insufficient, deficiency, and severe deficiency. This is crucial given the widespread prevalence of vitamin D deficiency and its impact on public health. Early detection can lead to timely interventions, improving bone health and overall well-being.

We use a non-invasive approach with machine learning techniques, specifically a hybrid CNN-ResNet model, to enhance prediction accuracy. The study analyzes a diverse dataset from 250 samples aged 18-30, focusing on various lifestyle factors like age, BMI, exercise, and sunlight exposure.

Performance metrics such as precision, recall, and F1 score are key to evaluating our models. Explainable AI tools like SHAP and LIME improve the model's interpretability, helping healthcare professionals make informed decisions. The expected outcomes include a valuable tool for personalized healthcare recommendations, enabling early identification of at-risk individuals and supporting preventive measures. Ultimately, this project seeks to improve health outcomes by providing targeted interventions for vitamin D deficiency across diverse populations.

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**A**

**Project Report on**

# "Vitamin D Deficiency Detection using Machine Learning"

submitted in partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY**
**IN**
**COMPUTER SCIENCE AND ENGINEERING**

Submitted by


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**K. Nishitha (VU21CSEN0100277)**

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**K. Tejaswi (VU21CSEN0101466)**


Under the esteemed guidance of


**Dr. Rita Roy M. Tech, Ph. D**
**Assistant Professor**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GITAM SCHOOL OF TECHNOLOGY

GITAM (Deemed to be University)

VISAKHAPATNAM

2024

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GITAM SCHOOL OF TECHNOLOGY

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# DECLARATION

I hereby declare that the project report entitled **"Vitamin D Deficiency Detection using Machine Learning"** is an original work done in the Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

Date:29/10/2024

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**GITAM (Deemed to be University)**

# CERTIFICATE

This is to certify that the project report entitled **"Vitamin D Deficiency Detection using Machine Learning"** is a bonafide record of work carried out G. Likhitha (VU21CSEN0100281), K. Nishitha (VU21CSEN0100277) , Ch. Chaitanya Sai (VU21CSEN0100392), K. Tejaswi (VU21CSEN0101466) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering.

Date :29/10/2024

Project Guide — Head of the Department

**Dr. Rita Roy** — **Dr. Lakshmeeswari**

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## Abstract .

Vitamin D deficiency is considered to be a global health disorder, and the need for a reliable, simple, and non-invasive prediction method is ever-rising. In this study, a survey was conducted that collected data from people belonging to Visakhapatnam, Hyderabad, and Maharashtra to gather a dataset of 250 samples from a group who are aged between 18 and 30 years. Independent variables include age, gender, weight, height, body mass index, waist circumference, percent body fat, bone mass, exercise frequency, exposure to sunlight, milk consumption, and vitamin D supplementation.

This research study has been aimed to predict the severity of VDD in a body. All the objectives aim to detect which set of different algorithms based on machine learning had the predictive efficiency more precise regarding Vitamin D Deficiency, which implies all the prime parameters, such as precision, recall, F1-score, accuracy, and AUC-ROC.

This paper will test a broad variety of machine learning algorithms; these are KNN, Decision Trees, Random Forest, BC, GB, SVMs, and MLPs. The study also looks into how much predictability might be added using a hybrid method that combines the best selection made between CNN and ResNet.

In addition to algorithmic assessment, this study looks at lifestyle and environmental contributors to Vitamin D status to identify significant predictors of VDD severity. Having patterns of exercise, dietary intake, and sunlight exposure evaluated, this research supports the idea that healthcare recommendations need to be tailored to promote habits that build Vitamin D. Techniques used: The use of SHAP and LIME are employed to improve the interpretability of machine learning models. In its contribution, the research contributes to public health efforts by providing accessible assessment tools coupled with awareness of the modifiable factors necessary for the maintenance of optimum levels of Vitamin D.

## Introduction

Vitamin D is a vital nutrient with extensive influence across numerous bodily functions, yet nearly one billion people worldwide are significantly affected by Vitamin D deficiency (VDD). Such deficiency tends to be linked with several autoimmune diseases like cardiovascular diseases, diabetes, and breast cancer. With the production of innumerable medical data every single day, while traditional methods of analysis are complicated and time-consuming, VDD in patients goes unnoticed. This study uses a combination of machine learning models, namely the hybrid CNN and ResNet algorithm, to predict the severity of VDD and classify cases into deficiency, insufficiency, and sufficiency categories.

Machine learning has gained much versatility and popularity with ease of use and high performance, especially in healthcare applications. Its ability to derive insights from data makes it widely adopted in healthcare applications. Machine learning models learn from training data and then use this knowledge to make predictions on new data, which is extremely valuable for the prediction of VDD severity. Unlike other studies which in some cases used much smaller datasets and solely relied on traditional statistical models, this study also takes into account a hybrid model, CNN and ResNet model, to increase predictive accuracy, given the bigger dataset. This hybrid model, inculcating deep learning capabilities of Convolutional Neural Networks along with the feature extraction strength of ResNet together, would be able to make more careful predictions by capturing complex data patterns. Then, the paper discusses variability in VDD etiology across different geographic and climatic contexts.

This study is designed with two main goals: (1) to predict the severity of VDD by implementing a wide range of machine learning models, such as Random Forest Classifier ,K-Nearest Neighbor, Decision Tree, Multilayer Perceptron, Cat Boost Classifier, LightGBM, Stochastic Gradient Classifier,Support Vector Machine, Gradient Boosting Classifier, and our hybrid CNN and ResNet model; and (2) to compare the classification performance of these models by implementing evaluation metrics in terms of Precision, Recall, F1-score, and ROC curves.

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Modern datasets are found to defy traditional methods: recent studies indicate that machine learning algorithms are more effective in handling such large datasets. It not only enhances the diagnostic capability but also proves new etiologies for VDD, thus making interventions more targeted in public health. The study proposed a hybrid CNN and ResNet model to obtain an inexpensive, accurate, and scalable form of assessment of the VDD severity level when compared with earlier studies that employed traditional models.

This work makes use of Explainable AI techniques: here, SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations) can be used to make improvements in the interpretability of models within machine learning. These techniques will permit better feature contribution to understand the critical factors making VDD severity worse or better. Therefore, integration of machine learning models- in particular the CNN and ResNet hybrid-preserves critical advancement in the prediction of VDD severity, filling an extremely important gap in search for more accessible efficient diagnostic tools. Ultimately, this research contributes to public health efforts through creating accessible assessment tools and making people aware of the modifiable factors ensuring optimal levels of Vitamin D.

## Literature Review

The literature review provides a synthesis of multiple studies that focus on Vitamin D deficiency detection using machine learning approaches.

| **Title** | **Authors and Year** | **Methodology Used** | **Advantages** | **Limitations** |
| --- | --- | --- | --- | --- |
| Vitamin D Deficiency Detection: A Novel Ensemble Approach with Interpretability Insights | Md Fahim,2024 | The system uses CatBoost and LightGBM models to predict Vitamin D deficiency with 96% accuracy and explains the important lifestyle factors using SHAP. | The approach effectively predicts Vitamin D deficiency with high accuracy and offers valuable insights into lifestyle factors, improving healthcare decisions and patient outcomes. | The study may be impacted by potential dataset biases affecting generalizability, a need for further clinical validation, ethical considerations, and possible resource constraints for real-time monitoring deployment. |
| Proposing an Effective Deep Learning Model for Vitamin D Deficiency Diagnosis | Asad Khattak,2024 | Utilizes convolutional neural networks (CNNs) to classify Vitamin D deficiency based on pre-processed patient lifestyle and demographic data. | The deep learning model offers improved accuracy in classifying Vitamin D deficiency compared to traditional methods, allowing for more reliable risk assessments and targeted health interventions. | The model's performance is limited by its reliance on a dataset that lacks direct clinical or biochemical data, which may affect generalizability and interpretability. |
| Leveraging Machine | Mrinal Saha,2023 | Data was collected from 50 SLE patients on | The study offers insights into the factors | The small sample size and the |

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| Learning to Evaluate Factors Influencing Vitamin D Insufficiency in SLE Patients | . | demographics and vitamin D levels, and analyzed using Linear Regression and Random Forest models to determine important factors. | affecting Vitamin D levels in SLE patients, potentially guiding personalized treatment plans and improving patient management strategies. | exclusion of significant factors, such as corticosteroid therapy and complement levels, limit the comprehensiveness and reliability of the findings. |
| Prediction of Vitamin D Deficiency in Older Adults: The Role of Machine Learning Models | John D Sluyter,2022 | The study involved 5,106 adults aged 50-84, randomly split into training and testing groups, and developed five machine learning models to predict serum vitamin D levels, evaluating their performance using metrics like AUC. | The research highlights the potential of machine learning models in predicting Vitamin D deficiency, which could aid in early detection and intervention strategies for older adults. | Accuracy is affected by subjective factors, regional differences, and missing important elements like fish consumption and genetics. |
| A Predictive Performance Analysis of Vitamin D Deficiency Severity Using Machine Learning Methods | G Sambasivam,2020 | Multiple machine learning models were tested for predicting Vitamin D deficiency, with Random Forest achieving the highest accuracy of 96% after data cleaning and feature selection. | The study provides a reliable model for predicting Vitamin D deficiency severity in young adults, which can guide targeted interventions and improve health | The dataset's focus on 18-21-year-old college students and its location-specific results may limit the applicability and validation of findings across |

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| | . | | outcomes in this demographic. | different age groups and climates. |
| A Novel Approach for Prediction of Vitamin D Status Using Support Vector Regression | Shuyu Guo,2013 | The study employed Random Forest (RF) and Radial Basis Function Support Vector Regression (RBF SVR) algorithms for modeling vitamin D levels. | The study demonstrates the effectiveness of machine learning algorithms in accurately modeling Vitamin D levels, potentially improving screening and management for individuals at risk of deficiency. | The study relied on a single validation dataset, struggled to predict extreme vitamin D levels, and may not accurately identify high-risk groups. |

These studies underscore the diverse approaches and promising results achieved in Vitamin D deficiency detection using machine learning. Nevertheless, they also highlight significant limitations, from dataset biases and location-specific outcomes to generalizability challenges across populations. Our study, integrating CNN and ResNet in a hybrid model, aims to address some of these gaps by leveraging the feature extraction capabilities of convolutional networks, ultimately advancing Vitamin D deficiency prediction with greater accuracy and reliability across varied populations.

## Problem Identification

Vitamin D deficiency (VDD) is a pervasive global health concern, affecting over one billion people. This deficiency, associated with serious autoimmune disorders like cardiovascular disease, diabetes, and various cancers, disproportionately impacts different populations, especially those with limited access to direct sunlight, dietary sources, and healthcare resources for screening and treatment. Traditional means of VDD identification, such as by biochemical assessment through serum 25(OH)D concentration testing, are invasive, costly, and sometimes inaccessible for routine screening or widespread populations. In addition, their basis on statistical models may be computationally intense in scale and not generalizable to larger populations or subpopulations.

With such limitations, along with rising demands for easy, quick, and available methods to predict and subsequently manage VDD, one realizes the urgent need to have new diagnostic approaches. ML advancements may now create the possibility to bridge that gap by predicting VDD severity using non-invasive features that are easy and accessible, such as age, BMI, waist circumference, dietary habits, and lifestyle. This shift from biochemical to data-driven prediction is expected to make detection more practical, affordable, and widely applicable, especially in low-resource settings or for individuals who are otherwise unable to access frequent healthcare assessments. Despite these advancements, a significant challenge persists in adapting these models for accuracy and reliability across various demographics and regions, as VDD is influenced by environmental and individual factors that can vary substantially.

Apart from leveraging the potential of machine learning models, this research focuses on the optimization of feature selection and data preprocessing for increasing the accuracy and interpretability of VDD predictions. In this research, through identification of the most influential non-invasive factors like sunlight exposure, levels of physical activity, and dietary habits, data dimensionality can be reduced, and overfitting can be minimized along with efficiency enhancement of models. The research will approach tackling demographic variability by iteratively testing various algorithms and hybrid architectures, including CNNs and ResNet. The model will be designed as adaptive to different populations and ultimately as a flexible framework that could easily be implemented in any healthcare system, in support of early intervention and personally customized care in managing VDD. The purpose of this paper is to overcome the problem by constructing a predictive model using machine learning algorithms in non-invasive

VDD detection and exploring the idea of using a hybrid approach between Convolutional Neural Networks (CNN) and ResNet to increase precision in classification. Ultimately, it will be contributing to building a reliable, scalable tool that healthcare providers may use in screening and managing VDD patients in various clinical settings.

## Objectives

- Tentative development of the different types of machine learning models for prediction in VDD severity: such as classifiers, especially KNN, DT, RF, CatBoost, Bagging Classifier, SGD, GB, SVM, and MLP classifiers that are selected for a distinct strength in the area of complexity and interaction between the data, thus it can enhance the robustness of the accuracy of predictions with different populations.
- Implement a hybrid deep learning model combining CNN and ResNet to leverage their strengths in deep feature extraction and hierarchical pattern recognition. This hybrid model will allow for more nuanced detection of VDD severity by capturing complex data interactions that traditional machine learning classifiers may miss, improving overall prediction quality.
- Assess your model's performance through such key performance metrics as Accuracy, Precision, Recall, F1-score, and ROC-AUC, which might give the best holistic presentation of model effectiveness for good sensitivities and specificities of VDD predictions.
- A Embed interpretability capabilities in the model using SHAP (SHapley Additive exPlanations) that will allow understanding the contribution of individual predictors, which include lifestyle habits, demographic factors, and body metrics. Better attention on model transparency by healthcare providers and researchers will ensure better understanding of how certain features are able to contribute to VDD predictions and support better decision-making in medicine.
- Validate model generalizability across diverse datasets to ensure its reliability in varying real-world settings, including the intended populations within Visakhapatnam, Hyderabad, and Maharashtra. This objective focuses on verifying the model’s adaptability and identifying any geographic or demographic limitations.

- Compare the hybrid CNN-ResNet with traditional models to establish superiority in predicting accuracy and in terms of processing efficiency for the model. Statistical model comparison will establish the added value of machine learning-based VDD detection, leading to adopting it into clinical workflow as a cost-effective and non-invasive procedure.
- This research will work towards producing an accurate and reliable tool for VDD detection, which allows health professionals to implement scalable solutions that deliver timely, accessible, and affordable health interventions to a range of clinical and community settings.

## Existing Model

VDD detection models by default are ensemble machine learning classifiers. These models collect information from multiple learning algorithms that are combined to augment predictive accuracy, reliability, and robustness. Other ensemble methods, such as RF, GB, and CatBoost, that primarily work on variable input features, which include demographic, lifestyle, and health parameters like BMI, sunlight exposure, and dietary habits, can be easily applied to patient records and structured datasets for a non-invasive approach in determining risk for VDD by considering easily accessible patient data.

Most ensemble models provide sufficiently good performance for a generalized large population by targeting at high-risk patients to potentially conduct further biochemical checks or provide preventive treatment. Much literature often reports impressive metrics such as accuracy, that could range from above 90%, for VDD classification if employing models like RF and GB, since it does strong feature set management as well as data-set-based correlation. Therefore, such ensemble models have emerged as benchmarks for non-invasive VDD detection, enhancing the efficacy of screening and allowing for early intervention, more importantly in low-resource environments.

Interpretation techniques such as SHAP (SHapley Additive exPlanations) often come in handy for model predictions and feature significance interpretation; these helps illustrate how much each predictor, such as sunlight exposure or BMI, contributes to the final prediction. Such interpretability features are particularly helpful in clinical settings, where understanding variable importance is critical for patient education and intervention planning.

**Technical Limitations and Drawbacks:**

This current VDD detection model would not generalize well across these diverse populations because of diverse environmental factors, diets, and genetic predispositions. Consequently, it calls for one to retrain and calibrate models with regional data that often proves to be resource-demanding.

**Limited Integration into Clinical Workflows:** The primary model of these models is developed as stand-alone applications, and therefore they cannot be very easily integrated into the electronic health record system or to a clinical decision support tool. This limits their real-time use within clinical environments where most of the benefits would be derived.

**Lack of Multimodal Data Analysis:** Structured Demographic and Lifestyle data form the basis for most VDD detection models, which limits their potential. For example, VDD assessments in clinical settings require multimodal data, including medical images, blood reports, and genomic data. Such multimodal inputs cannot be handled or analyzed by existing models and, as such, restrict their applicability for comprehensive VDD risk assessment.

**Dependency on Structured Data:** The ensemble models require rightly formatted data inputs for them to preserve their quality in making good predictions. Missing or unaligned data fields might thus compromise the accuracy, thereby making it challenging to implement these models in under-resourced clinics or health centers with rather archaic systems of managing patient data.

**Computational Constraints:** Ensemble models are computationally less intensive in nature as compared to complicated deep learning models. However, they still need a minimum threshold of computational resources for efficient operation. Such a need can be a constraint in low-resource environments wherein limited technology infrastructure may restrain the application of a model.

**Challenges with Real-Time Application:** Predictive models for VDD are commonly designed on a batch processing paradigm. This can create limited usability when these are utilized within real-time settings. Thus, such models could not serve as much use within the clinical's quest to be quicker and make decisions within minutes clinically.

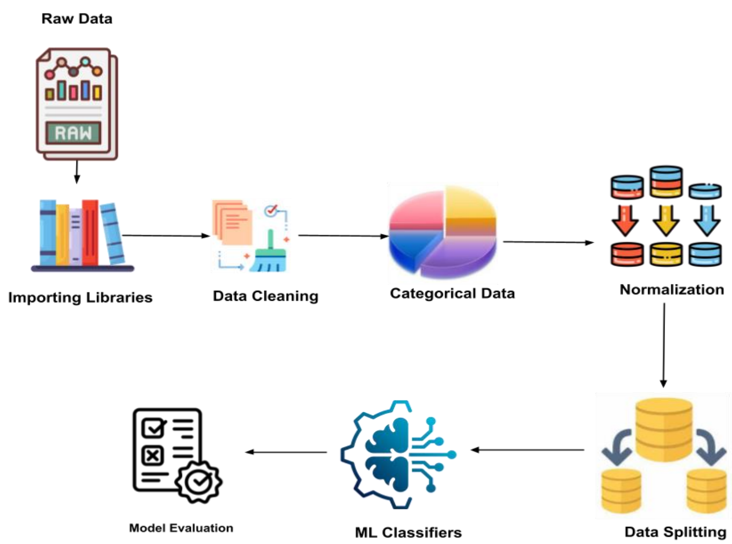
**Accuracy Variation in Extreme Cases :** While ensemble models work very well on most cases, they are likely to be less accurate for extreme cases of VDD, especially when the patient has some unique health or genetic factors. This is usually a problem for those patients with complex health conditions or unusual metabolic profiles, who therefore need some specific diagnostic criteria.

We focus on overcoming the limitations mentioned above in our study through a hybrid model of CNN and ResNet to improve predictive capabilities. This hybrid model will overcome some problems facing traditional ensemble models.

## Proposed System

The Vitamin D Deficiency Detection System is a machine learning system that can identify the patient's risk of having Vitamin D deficiency based on patients' lifestyle habits. Based on data regarding diet, sunlight exposure, physical activities, and other relevant habits, the system provides accurate Vitamin D deficiency risk determination to a healthcare professional and does not require clinical notes or lab test results.

The model, using techniques like SHAP and LIME in Explainable AI, ensures that the predictions are interpretable; this provides clear insights for healthcare providers regarding lifestyle factors that increase deficiency risk. The system's interpretability brings forth trust and helps support professionals in making data-driven recommendations. Since it allows for real-time adaptability, it can adapt with new research in lifestyle influences and Vitamin D, and its predictions will be entirely accurate and up-to-date. Its scalability allows it to be widely dispersed across varied healthcare environments.




**Fig(1). Proposed system**

**Key features include:**

1. Lifestyle-Based Deficiency Detection : the model focuses strictly on the lifestyle habits of patients, which indicate who is at risk and encourages early interventions without the involvement of clinical tests or notes.

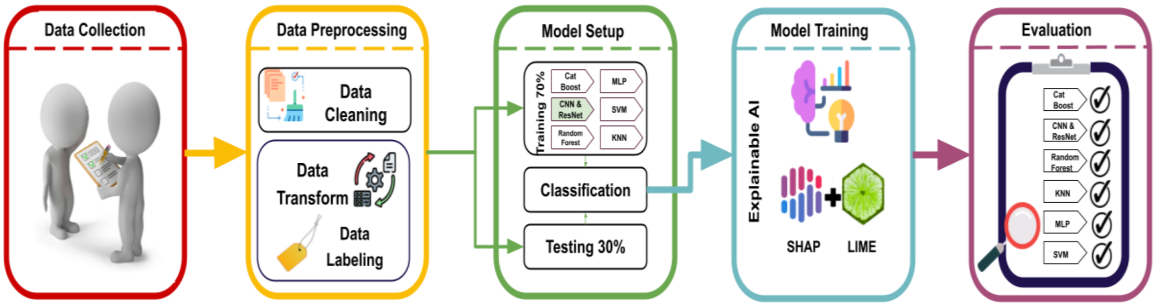
2. Explainable AI for Transparency : SHAP and LIME are interactively explaining the model's predictions, which will provide provider analysis of what factors underlie every deficiency risk assessment.

3. Personalized Recommendations : The system gives the patient recommendations that are tailored to their needs, like dietary changes or modifications in outdoor activities to regulate their Vitamin D levels.

4. Real-Time Learning The model learns about Vitamin D deficiency in lifestyle-related activities. Thus, this model remains relevant and accurate at the time of assessment without requiring full-time retraining.

This Vitamin D Deficiency Detection System offers a more practical and data-driven approach to preventive care, focusing on lifestyle data to reduce Vitamin D deficiency through informed and actionable insights.

## System Architecture




**Fig(2). Architecture of the model**

The architecture for the Vitamin D Deficiency (VDD) severity prediction system is being developed in three main stages with several components currently under development:

DataAcquisition:

• 250 samples were collected based on demographic and lifestyle characteristics, which include age, BMI, exercise, time spent exposed to sunlight, and the consumption of dairy products.

• The data were collected from different places in districts of Maharashtra, Telangana, and Andhra Pradesh.

Data Preprocessing:

• Data Cleaning:

- ➢ Missing values were appropriately imputed to preserve the integrity of the datasets.
- ➢ Outliers were identified and corrected for because they may determine the outcome of the final results.
- ➢ The dataset was standardized to make sure the features have a uniform standard across them.
- ➢ Data Transformation: Continuous variables were normalized and the categories were encoded, this ensures that the data is prepared for model input

• Data Balancing:

That which addressed class imbalances in case they occurred in the datasets were applied.

Machine Learning Model Implementation:

- A hybrid CNN-ResNet algorithm is trialed with other models for showing results. The tested models in this study include the following KNN, DT, RF, LightGBM, CatBoost (CB), Bagging Classifier (BC), Gradient Boosting (GB), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). It aims at classifying the severity of VDD.
- The system is currently in the tuning stage where hyperparameters and cross-validation are used to optimize how well the model works.
- Preliminary tests show a high potential for the hybrid CNN-ResNet model. All models were evaluated with metrics like Precision, Recall, F1-Score, and AUC.

Explanable AI:

- For easy understanding, the system is enhanced with SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). This explains which features are important, how the model comes to a prediction, and increases transparency and trust in the results.

## Tools/Technologies Used

### Software Requirements

1. Programming Language:

- Python is chosen because it boasts a large collection of tools, is easy to read, and has good community help in data science and machine learning. The best thing is to use version 3.8 or higher because it works well with the latest features and updates in the AI libraries.
- Python Advantage: Large libraries and structured code make it a great language for building things piece by piece. It is helpful in rapid prototyping and, therefore, in rapid model testing and validation.

2. Libraries:.

- Scikit-learn: An important set of tools for machine learning techniques, which include methods for classification, regression, clustering, and ways to check model performance. It is also very fast for first-round development and testing of models.
- XGBoost, LightGBM, CatBoost: Each of these boosting libraries has special merits:
  - XGBoost: Particularly strong for tabular data and offers enormous flexibility.
  - LightGBM: Designed to be fast and efficient, it can handle large datasets and complex data well.
  - CatBoost: Can handle categorical data directly, which makes it great for datasets with many categorical features.
- TensorFlow/Keras: These tools help build complex neural networks (including CNNs and RNNs) with user-friendly wrappers (like Keras) and make massive use of the GPU for parallel processing, making training much faster.
- SHAP: Helps in understanding how well the models are working by showing the contribution of each feature to the result.
- LIME: By slightly changing data and watching prediction behavior, it gives insight into how predictions should be understood. All the above tools make it easier to trust this model and understand its logic.
- Pandas and NumPy: Pandas is essential for data handling, supporting filtering, grouping, and manipulation of data. NumPy performs efficient numerical operations as well as array handling on high-dimensional data.
- Matplotlib or Seaborn: While Matplotlib handles low-level, highly customizable plotting, Seaborn is designed for complex statistical plotting with fewer lines of code. Seaborn also enables effective data visualization.

3. Development Environment:

- Jupyter Notebook/Anaconda: Jupyter allows for interactive and sequential code execution, ideal for testing and visualizing results. Anaconda manages environments and dependencies, simplifying library installation and version management.
- Resource Addition - Git: Enables tracking of changes made in the code, making cooperation and updates throughout development easier.

**Hardware Requirements**

1. Processor:

- Intel i5 or better or AMD Ryzen 5 or better: This will ensure sufficient processing power for complex calculations at both the data processing and model training stages.
- Parallel Processing: Best suited for multi-core processors or a multi-threaded CPU, as many libraries like Scikit-learn and TensorFlow support parallel processing, which can improve performance.

2. RAM:

- 16GB+: Sufficient RAM ensures that large datasets can be processed without crashing, enabling efficient data handling and faster model training.
- Expandable RAM: Devices with expansion slots for additional RAM are preferable, as memory can be increased later to handle big data or more complex models in the future.

3. Storage:

- 512GB SSD or greater: Fast storage speeds up loading, saving, and retrieving data, making it easier to work with large files or load libraries quickly.
- Cloud Storage Option: Utilizing cloud storage, such as AWS S3 or Google Cloud Storage, for large datasets can help minimize dependence on local storage and provide data access flexibility.

## Conclusion

Our project addresses a major healthcare requirement by developing a model that predicts and categorizes the population into four groups in accordance with their vitamin D levels that are sufficient, insufficient, deficiency, and severe deficiency. It means a large number of patients lack vitamin D in significant amounts, and it actually influences public health. Vitamin D is very essential for bone health, the immune system, and overall well-being. Early discovery of deficiency can avert quick action to minimize related health risks.

A gentle and easy approach to finding problems early is used by the project through machine learning techniques. The study seeks to try to improve on the accuracy and trustworthiness of the predictions as well as learn more about what causes vitamin D deficiency through different kinds of algorithms, including a mix of CNN and ResNet models. This applies a specific dataset, culled from 250 subjects across the age range of 18-30. Factors within their lifestyle are including these, namely age, gender, BMI, waist size, percentage body fat, physical exercise, sun exposure, consumption of milk, and medicines. This would relate on how these factors impinge and work in effect to cause effects on Vitamin D. The more explicit detail provided gives more specific health care to be done.

Other performance metrics like precision, recall, accuracy, and F1 score will form part of how the project will be graded. The study is set to research if other models of machine learning can effectively identify which one is best. The use of AI by the aid of tools in Explainable AI such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) gives insight into what the model is predicting. This will enable building trust among the health personnel. It gives a reason to the health workers about why the model yields particular results. Therefore, there will be better decisions relating to care for patients.

Although this project is still in development, the expected outcomes show that it will create a useful tool for personalized healthcare advice. This model gives information about a person's vitamin D levels and helps with plans to prevent deficiencies. By finding these at-risk people early, healthcare providers can suggest changes to their lifestyle, diet, or supplements that fit each person's needs.

This will also open an avenue for rethinking deficiency like vitamin D screening and management, hence improving population health outcomes. In the course of the project, information gathered is expected to be scientifically useful not only in enhancing understanding about vitamin D deficiency but also in empowering healthcare systems to implement targeted interventions for better health for everyone. Goal: The development of a strong monitoring and control mechanism of vitamin D levels among different population groups to ensure easy access to effective means of prevention.

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Guide: Dr. Rita Roy
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# Vitamin D Deficiency Detection Using Machine Learning

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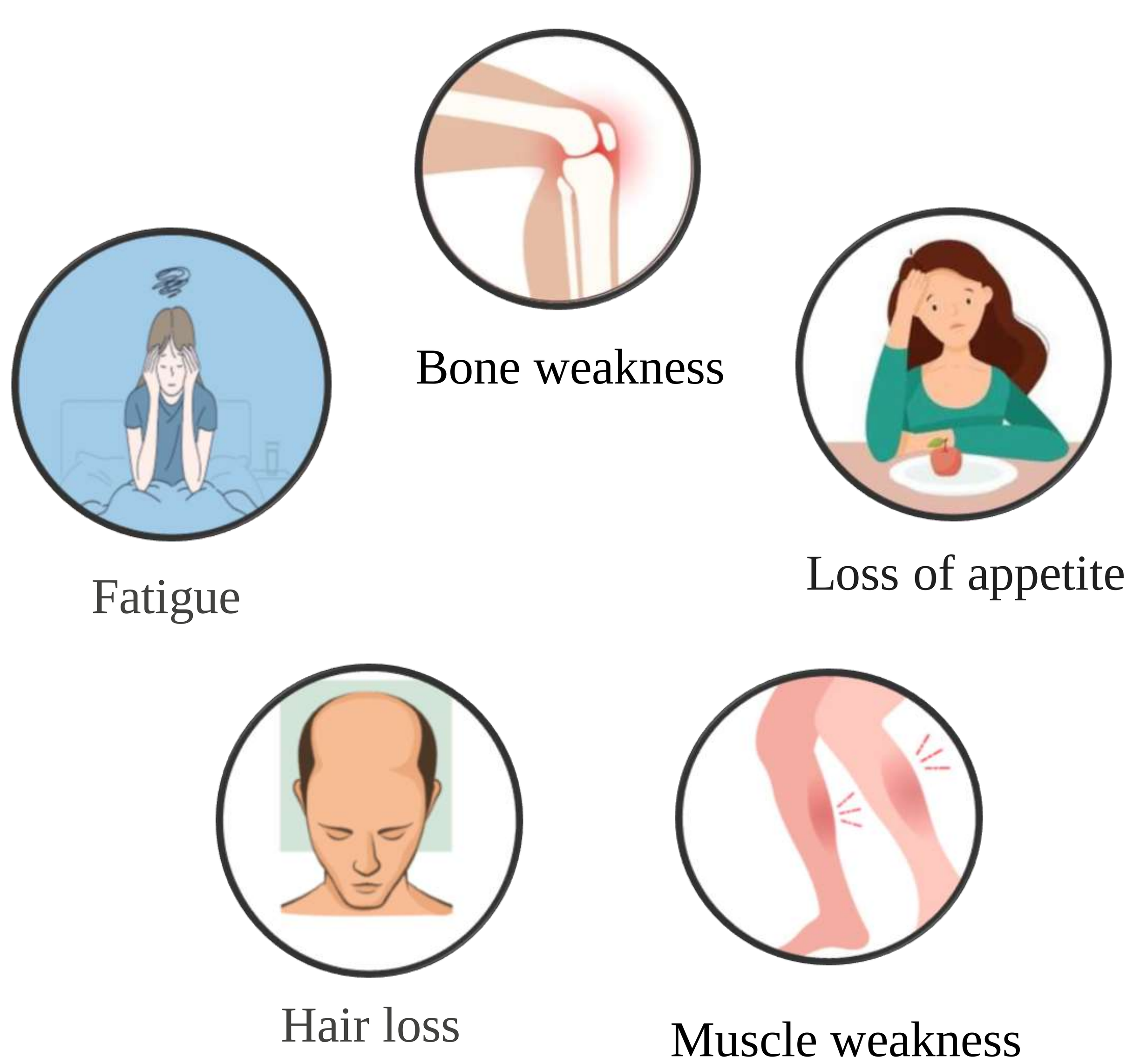
## ABSTRACT

Vitamin D deficiency (VDD) has become a global health concern, creating a need for reliable, non-invasive prediction methods. This project analyzes serum Vitamin D levels in 250 individuals (aged 18-30) from Visakhapatnam, Hyderabad, and Maharashtra, examining lifestyle and environmental factors like BMI, exercise frequency, and sunlight exposure. Machine learning models—including KNN, Decision Tree, Random Forest, and a CNN-ResNet hybrid—are evaluated using metrics like Precision, Recall, F1-score, and AUC-ROC for validating classifier effectiveness. Explainable AI tools (SHAP, LIME) further highlight predictors impacting VDD, supporting personalized healthcare recommendations and public health initiatives.

## INTRODUCTION

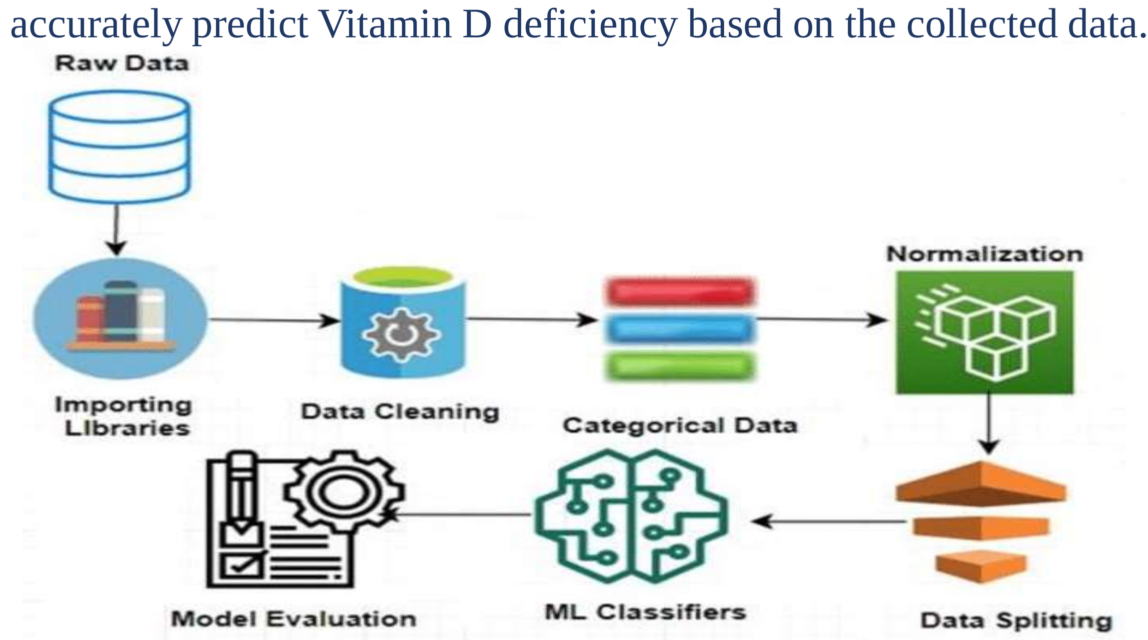
Vitamin D is essential for numerous bodily functions, yet nearly a billion people worldwide are impacted by Vitamin D deficiency (VDD), which is linked to autoimmune conditions, including cardiovascular disease, diabetes, and cancers like breast cancer. Traditional diagnostic methods struggle with accuracy and scalability, making it challenging to effectively identify and categorize VDD. In response, this project applies advanced machine learning models, including a hybrid CNN and ResNet, to predict VDD severity and classify cases into deficiency, insufficiency, and sufficiency categories. Machine learning models like Random Forest, Decision Tree, and the hybrid CNN-ResNet provide nuanced insights, leveraging large datasets to achieve higher diagnostic accuracy and efficiency. The project further incorporates Explainable AI techniques, SHAP and LIME, to interpret model predictions and reveal critical factors influencing VDD. This research ultimately supports public health by offering a scalable, cost-effective, and interpretable tool for proactive VDD assessment and promoting modifiable habits essential for optimal Vitamin D levels.

## SYMPTOMS

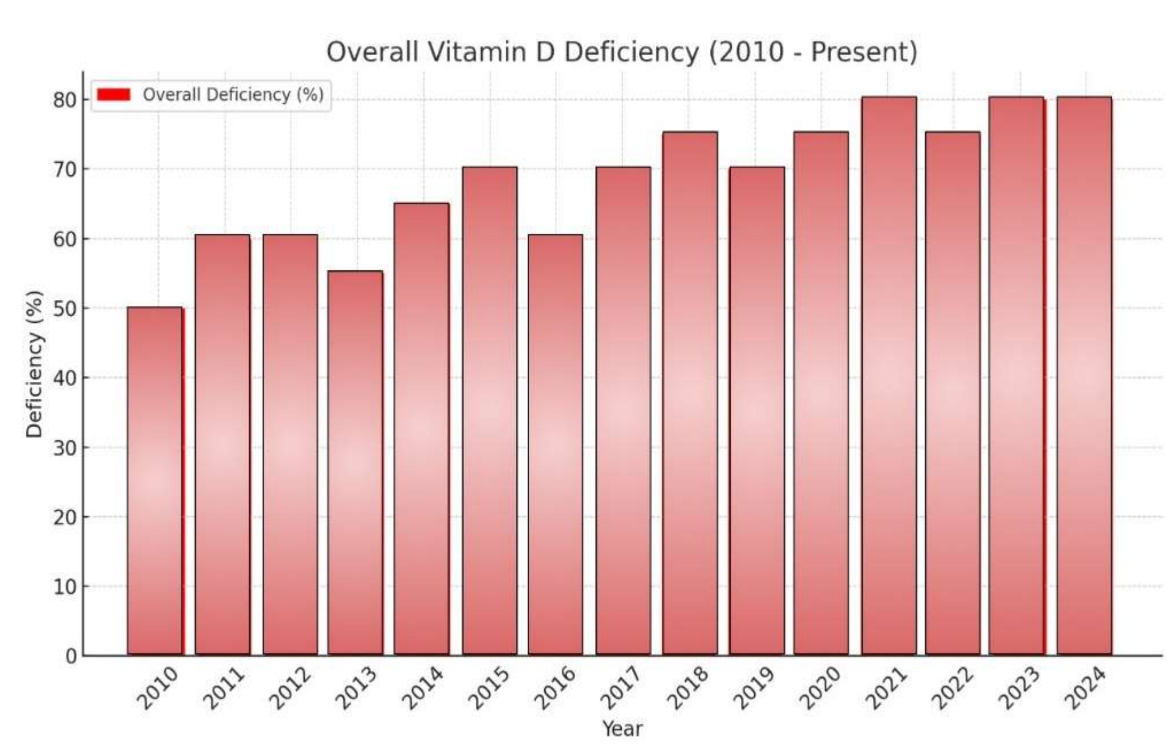



## METHODS

Data was collected from young adults (aged 18-30) through structured surveys, capturing variables such as age, sex, BMI, exercise habits, sunlight exposure, and medications. Various machine learning models, including Random Forest, K-Nearest Neighbors (KNN), Decision Trees, LightGBM, and CatBoost, were employed for analysis. Random Forest improves accuracy by aggregating multiple decision trees, while KNN classifies data based on proximity to neighbors. The dataset was split into training (70%) and testing (30%) subsets, with k-fold cross-validation applied for robustness. Model performance was evaluated using metrics like accuracy, precision, recall, and F1 score. Ethical considerations included ensuring participant data privacy and compliance with ethical guidelines. This approach aims to accurately predict Vitamin D deficiency based on the collected data.




## OVERALL VITAMIN D DEFICIENCY (2010 – PRESENT)




## DATA ANALYSIS

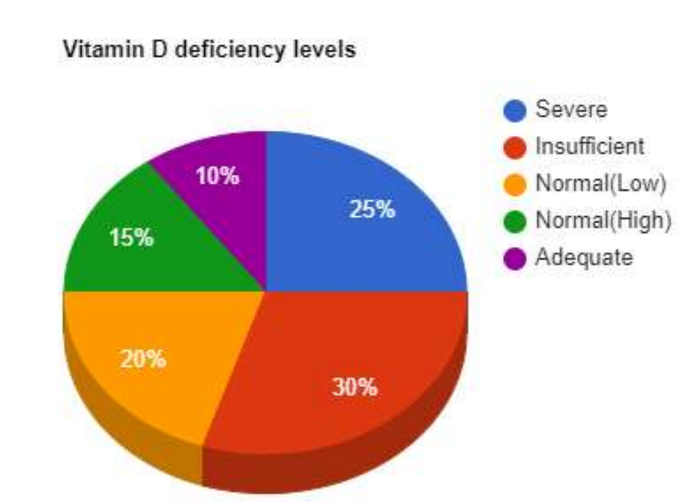
Each machine learning model's performance was evaluated using key metrics: accuracy to measure overall correctness, precision for the reliability of positive predictions, recall to capture sensitivity to true positives, and F1-score to balance precision and recall. ROC curves further assessed each model's ability to distinguish between classes, providing a comprehensive view of predictive performance.

Cross-validation techniques were employed to test model robustness, ensuring the models' reliability across different data subsets. This method enhanced the evaluation by verifying consistency in performance, helping identify the best-suited model for predicting Vitamin D deficiency severity.

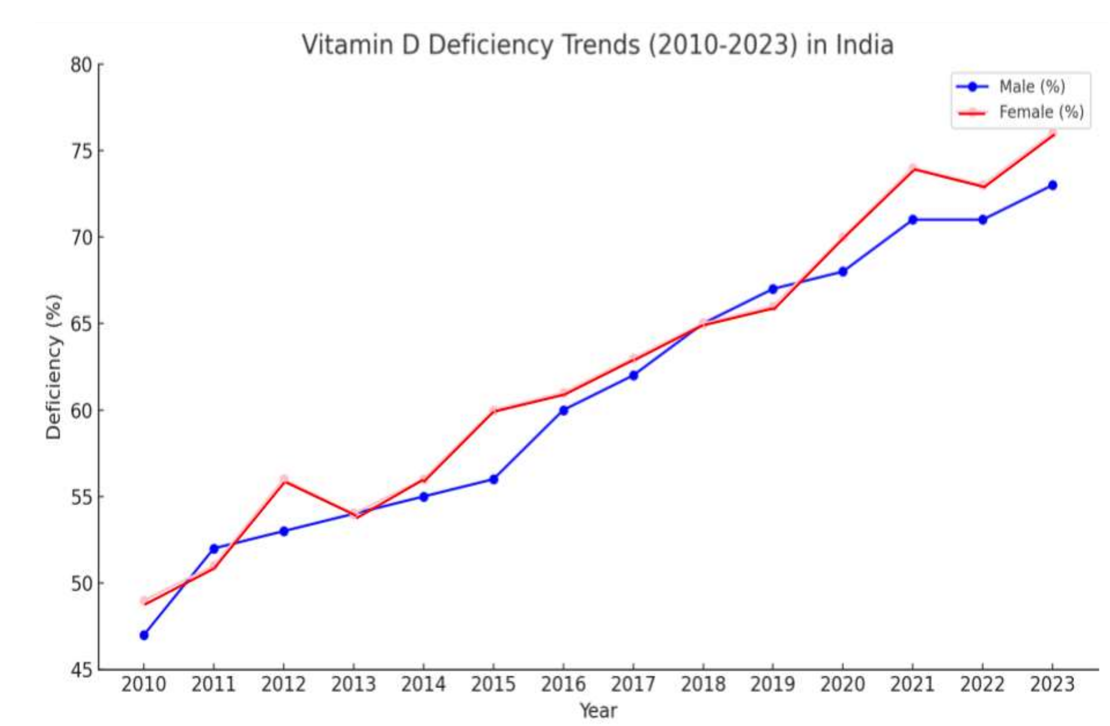
## OBJECTIVES

- This model will harness the feature extraction capabilities of Convolutional Neural Networks (CNN) alongside the residual learning framework of ResNet to enhance the detection of VDD severity.
- A range of performance metrics will be utilized to provide a comprehensive assessment of the model's effectiveness, ensuring a balanced evaluation of its sensitivity and specificity in predicting VDD.
- By employing SHAP, we will gain valuable insights into how various predictors, such as lifestyle habits and demographic factors, influence the model's predictions, thereby enhancing interpretability.
- The model will be tested across diverse datasets to ensure its reliability and adaptability in different geographic and demographic settings, allowing for broader application.
- A comparative analysis will demonstrate the advantages of the hybrid model in terms of both prediction accuracy and computational efficiency, highlighting its effectiveness for VDD detection.
- The project will focus on how the model's predictions can be utilized in community health programs to raise awareness about Vitamin D deficiency and promote preventive measures. By collaborating with public health organizations, we aim to facilitate the implementation of targeted interventions that can improve Vitamin D status in the population.

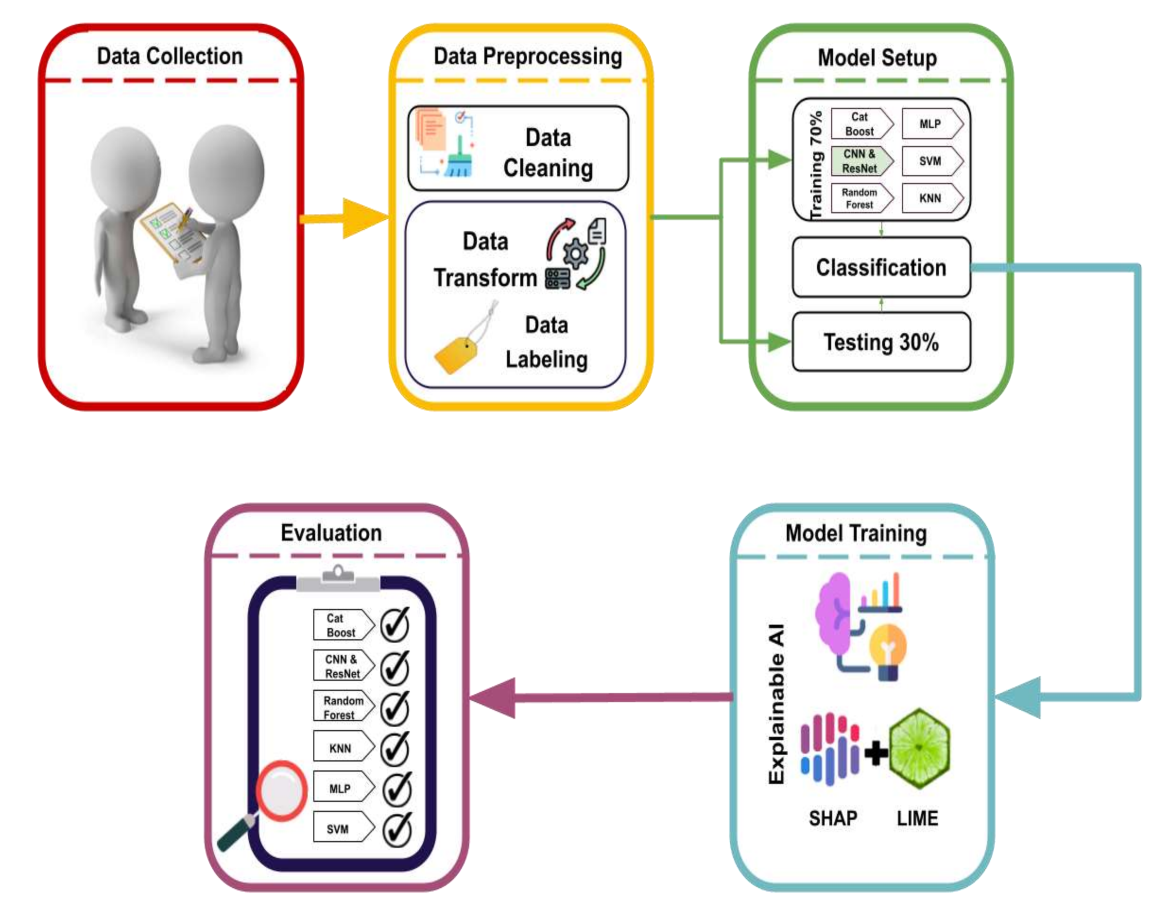
## POPULATION BREAKDOWN BY VITAMIN D STATUS




## VITAMIN D DEFICIENCY (2010 – 2023) IN INDIA




## SYSTEM ARCHITECTURE




## CONCLUSION

This study presents a hybrid deep learning model that integrates Convolutional Neural Networks (CNN) and ResNet to accurately detect Vitamin D deficiency (VDD) severity. By utilizing advanced machine learning techniques, we enhance predictive capabilities while ensuring interpretability through SHAP. Comprehensive evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC will be employed to assess model performance rigorously. Validation across diverse datasets will ensure reliability and adaptability in various contexts. This project aims to demonstrate the superiority of the hybrid approach over traditional models, positioning it as a valuable tool for public health improvement. Ultimately, our work seeks to contribute to innovative solutions for enhancing health and wellness in diverse populations.

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